Introduction to Multi-level Models

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
```

```
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

This chapter adapts David Rocke[1]'s May 25 – June 1 lectures from the 2021 edition of this course[2] (Rocke 2021).

1 Correlated data

The models in earlier chapters assumed that the outcomes $Y_1, \ldots, Y_n$ were mutually independent given the covariates. Many epidemiologic studies collect data in **clusters**: repeated measurements on the same person, patients treated at the same hospital, or houses in the same county (Dobson and Barnett 2018, chap. 11). Observations in the same cluster tend to resemble one another more than observations in different clusters, so the independence assumption fails. Ignoring that correlation usually makes standard errors too small, because the data carry less information than the same number of independent observations would (Dobson and Barnett 2018, chap. 11).

Two kinds of model term describe clusters. A **fixed effect** is a parameter for a specific group whose particular value we want to estimate, such as the effect of a specific treatment. A **random effect** treats the groups in the data as a sample from a larger population of groups, and models each group's effect as a random variable drawn from a distribution; the parameters of interest are then the parameters of that distribution, especially its variance (Rocke 2021).

Exm

Example 1.1 (Fixed versus random effects)**.**

- In a trial comparing a new drug with a placebo, the treatment groups are the only two groups of interest, so treatment is a fixed effect.
- In a study that measures each of 20 patients several times, the 20 patients are a sample from the population of all patients, and we want conclusions that apply to patients

[1] https://dmrocke.ucdavis.edu/
[2] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-Spring-2021.html

beyond these 20, so patient is a random effect.

Random effects can be **nested** or **crossed**. Effects are nested when each lower-level group belongs to exactly one higher-level group, such as batches within laboratories. Effects are crossed when every level of one factor can appear with every level of the other, such as patients and raters when every rater scores every patient (Rocke 2021).

A model that contains both fixed and random effects is called a **mixed model**. The same ideas extend from linear models to logistic and Poisson regression (Rocke 2021).

2 One-way random-effects models

The simplest model with a random effect has one grouping factor and no covariates (Rocke 2021; Dobson and Barnett 2018, chap. 11).

Definition 2.1 (One-way random-effects model)**.** Suppose we observe m outcomes $Y_{i1}, \dots, Y_{im}$ in each of a groups $i = 1, \dots, a$. The **one-way random-effects model** is

$$Y_{ij} \overset{\text{def}}{=} \mu + \alpha_i + \epsilon_{ij}, \tag{1}$$

where:

- μ is a fixed overall mean;
- $\alpha_1, \dots, \alpha_a \sim_{\text{iid}} \mathrm{N}(0, \sigma_\alpha^2)$ are the **random effects** of the groups;
- $\epsilon_{ij} \sim_{\text{iid}} \mathrm{N}(0, \sigma^2)$ are the within-group errors;
- the α_is and the ϵ_{ij}s are mutually independent.

The variances σ_α^2 and σ^2 are called **variance components**.

Exm

Example 2.1 (Batches of dyestuff)**.** The `Dyestuff` data in the `lme4` package (Bates et al. 2015) record the yield of dyestuff (in grams of standard color) from $m = 5$ preparations in each of $a = 6$ batches of an intermediate product. The batches are a sample of all the batches the process could produce, so a one-way random-effects model treats α_i as the deviation of batch i's mean yield from the overall mean μ.

```
with(Dyestuff, tapply(Yield, Batch, mean))
#>    A    B    C    D    E    F
#> 1505 1528 1564 1498 1600 1470
```

The shared random effect α_i is what makes observations in the same group correlated.

Theorem 2.1 (Moments of the one-way random-effects model)**.** *Under Definition 2.1, for any group i and observations $j \neq k$:*

$$\begin{aligned} \mathrm{E}[Y_{ij}] &= \mu \\ \mathrm{Var}(Y_{ij}) &= \sigma_\alpha^2 + \sigma^2 \\ \mathrm{Cov}(Y_{ij}, Y_{ik}) &= \sigma_\alpha^2 \end{aligned} \tag{2}$$

and observations in different groups are independent.

Proof

Proof. This proof uses the definition of covariance[a], the covariance of a variable with itself[b], and linearity of expectation.
Mean:

$$\begin{aligned} \mathrm{E}\big[Y_{ij}\big] &= \mathrm{E}\big[\mu + \alpha_i + \epsilon_{ij}\big] && \text{(definition of the model)} \\ &= \mu + \mathrm{E}[\alpha_i] + \mathrm{E}\big[\epsilon_{ij}\big] && \text{(linearity of expectation)} \\ &= \mu + 0 + 0 && \text{(both have mean zero)} \\ &= \mu && \text{(simplify)} \end{aligned} \tag{3}$$

Covariance, for j and k equal or not:

$$\begin{aligned} \mathrm{Cov}\big(Y_{ij}, Y_{ik}\big) &\overset{\text{def}}{=} \mathrm{E}\big[(Y_{ij} - \mu)(Y_{ik} - \mu)\big] && \text{(definition of covariance)} \\ &= \mathrm{E}\big[(\alpha_i + \epsilon_{ij})(\alpha_i + \epsilon_{ik})\big] && \text{(substitute the model)} \\ &= \mathrm{E}\big[\alpha_i^2 + \alpha_i\epsilon_{ik} + \epsilon_{ij}\alpha_i + \epsilon_{ij}\epsilon_{ik}\big] && \text{(distribute)} \\ &= \mathrm{E}[\alpha_i^2] + \mathrm{E}[\alpha_i\epsilon_{ik}] + \mathrm{E}\big[\epsilon_{ij}\alpha_i\big] + \mathrm{E}\big[\epsilon_{ij}\epsilon_{ik}\big] && \text{(linearity of expectation)} \\ &= \sigma_\alpha^2 + \mathrm{E}[\alpha_i\epsilon_{ik}] + \mathrm{E}\big[\epsilon_{ij}\alpha_i\big] + \mathrm{E}\big[\epsilon_{ij}\epsilon_{ik}\big] && (\mathrm{E}[\alpha_i] = 0, \text{ so } \mathrm{E}[\alpha_i^2] = \mathrm{Var}(\alpha_i)) \\ &= \sigma_\alpha^2 + 0 + 0 + \mathrm{E}\big[\epsilon_{ij}\epsilon_{ik}\big] && \text{(independent, mean-zero factors)} \end{aligned} \tag{4}$$

If $j \neq k$, ϵ_{ij} and ϵ_{ik} are independent with mean zero, so $\mathrm{E}\big[\epsilon_{ij}\epsilon_{ik}\big] = 0$ and the covariance is σ_α^2. If $j = k$, $\mathrm{E}\big[\epsilon_{ij}^2\big] = \sigma^2$, so the covariance of Y_{ij} with itself, which is its variance, is $\sigma_\alpha^2 + \sigma^2$. Observations in different groups $i \neq i'$ are functions of the disjoint sets of independent variables $\{\alpha_i, \epsilon_{ij}\}$ and $\{\alpha_{i'}, \epsilon_{i'k}\}$, so they are independent. □

[a] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov
[b] https://morrison-lab.github.io/pds/variance-covariance.html#lem-cov-xx

Exm

Example 2.2 (Covariance between two dyestuff preparations)**.** Suppose the batch-level variance in Example 2.1 is $\sigma_\alpha^2 = 1764$ and the within-batch variance is $\sigma^2 = 2451$. Then by Theorem 2.1 two preparations from the same batch would have covariance 1764, and each preparation would have variance $1764 + 2451 = 4215$. Two preparations from different batches would have covariance 0.

Definition 2.2 (Intraclass correlation)**.** Under Definition 2.1, the **intraclass correlation** is the correlation between two observations $j \neq k$ in the same group:

$$\rho \overset{\text{def}}{=} \frac{\mathrm{Cov}\big(Y_{ij}, Y_{ik}\big)}{\sqrt{\mathrm{Var}\big(Y_{ij}\big)\,\mathrm{Var}\big(Y_{ik}\big)}} \tag{5}$$

Corollary 2.1 (Intraclass correlation in terms of variance components)**.** *Under Definition 2.1,*

$$\rho = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2}. \tag{6}$$

Proof

Proof. This proof uses Theorem 2.1.

$$
\begin{aligned}
\rho &\overset{\text{def}}{=} \frac{\operatorname{Cov}(Y_{ij}, Y_{ik})}{\sqrt{\operatorname{Var}(Y_{ij})\operatorname{Var}(Y_{ik})}} && \text{(definition of intraclass correlation)} \\
&= \frac{\sigma_\alpha^2}{\sqrt{\operatorname{Var}(Y_{ij})\operatorname{Var}(Y_{ik})}} && \text{(covariance from the moments theorem)} \\
&= \frac{\sigma_\alpha^2}{\sqrt{(\sigma_\alpha^2+\sigma^2)(\sigma_\alpha^2+\sigma^2)}} && \text{(variances from the moments theorem)} \\
&= \frac{\sigma_\alpha^2}{\sigma_\alpha^2+\sigma^2} && \text{(square root of a square)}
\end{aligned} \tag{7}
$$

□

Exm

Example 2.3 (Intraclass correlation for the dyestuff batches)**.** With the variance components from Example 2.2, Corollary 2.1 gives

$$
\begin{aligned}
\rho &= \frac{1764}{1764+2451} && \text{(substitute the variance components)} \\
&= \frac{1764}{4215} && \text{(add)} \\
&\approx 0.4185 && \text{(divide)} \\
&\approx 0.42 && \text{(round to two decimal places)}
\end{aligned} \tag{8}
$$

so about 42% of the variation in yield is between batches.

The intraclass correlation is also the fraction of the total variance $\operatorname{Var}(Y_{ij}) = \sigma_\alpha^2 + \sigma^2$ that comes from the random effect.

3 Variance components

One way to estimate the variance components uses the mean squares from a one-way analysis of variance (ANOVA) table (Rocke 2021). Write $\bar{Y}_{i\cdot} \overset{\text{def}}{=} \frac{1}{m}\sum_{j=1}^{m} Y_{ij}$ for the mean of group i and $\bar{Y}_{\cdot\cdot} \overset{\text{def}}{=} \frac{1}{a}\sum_{i=1}^{a} \bar{Y}_{i\cdot}$ for the overall mean. The **between-group mean square** and the **within-group mean square** are

$$
\begin{aligned}
\text{MSA} &\overset{\text{def}}{=} \frac{m}{a-1}\sum_{i=1}^{a}(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2, \\
\text{MSE} &\overset{\text{def}}{=} \frac{1}{a(m-1)}\sum_{i=1}^{a}\sum_{j=1}^{m}(Y_{ij}-\bar{Y}_{i\cdot})^2.
\end{aligned} \tag{9}
$$

Theorem 3.1 (Expected mean squares in the one-way random-effects model)**.** *Under Definition 2.1:*

$$
\begin{aligned}
\mathrm{E}[MSE] &= \sigma^2 \\
\mathrm{E}[MSA] &= m\sigma_\alpha^2 + \sigma^2
\end{aligned} \tag{10}
$$

Proof

Proof. This proof uses the fact that the sample variance[a] of independent, identically distributed observations is unbiased for their variance, the law of iterated expectations[b], and the variance of a linear combination[c].

Within groups. Given α_i, the observations $Y_{i1}, \dots, Y_{im}$ are independent with common variance σ^2, so their sample variance $s_i^2 \overset{\text{def}}{=} \frac{1}{m-1}\sum_{j=1}^{m}(Y_{ij} - \bar{Y}_{i\cdot})^2$ satisfies $\mathrm{E}[s_i^2 \mid \alpha_i] = \sigma^2$. By the law of iterated expectations, $\mathrm{E}[s_i^2] = \sigma^2$. Then

$$\begin{aligned}
\mathrm{E}[\mathrm{MSE}] &= \mathrm{E}\left[\frac{1}{a}\sum_{i=1}^{a} s_i^2\right] && \text{(MSE is the average of the } s_i^2\text{)} \\
&= \frac{1}{a}\sum_{i=1}^{a}\mathrm{E}[s_i^2] && \text{(linearity of expectation)} \\
&= \frac{1}{a}\sum_{i=1}^{a}\sigma^2 && \text{(each } s_i^2 \text{ is unbiased)} \\
&= \sigma^2 && \text{(simplify)}
\end{aligned} \tag{11}$$

Between groups. Each group mean is $\bar{Y}_{i\cdot} = \mu + \alpha_i + \bar{\epsilon}_{i\cdot}$, where $\bar{\epsilon}_{i\cdot} \overset{\text{def}}{=} \frac{1}{m}\sum_{j=1}^{m}\epsilon_{ij}$. The group means are independent and identically distributed across groups, with

$$\begin{aligned}
\mathrm{Var}(\bar{Y}_{i\cdot}) &= \mathrm{Var}(\alpha_i) + \mathrm{Var}(\bar{\epsilon}_{i\cdot}) && \text{(variance of a sum of independent terms)} \\
&= \sigma_\alpha^2 + \frac{1}{m^2}\sum_{j=1}^{m}\mathrm{Var}(\epsilon_{ij}) && \text{(variance of a linear combination)} \\
&= \sigma_\alpha^2 + \frac{\sigma^2}{m} && \text{(simplify)}
\end{aligned} \tag{12}$$

MSA is m times the sample variance of $\bar{Y}_{1\cdot}, \dots, \bar{Y}_{a\cdot}$, so

$$\begin{aligned}
\mathrm{E}[\mathrm{MSA}] &= m \cdot \mathrm{Var}(\bar{Y}_{i\cdot}) && \text{(sample variance is unbiased)} \\
&= m\left(\sigma_\alpha^2 + \frac{\sigma^2}{m}\right) && \text{(substitute)} \\
&= m\sigma_\alpha^2 + \sigma^2 && \text{(distribute)}
\end{aligned} \tag{13}$$

□

[a] https://morrison-lab.github.io/sds/exploratory-descriptive.html#def-sample-variance
[b] https://morrison-lab.github.io/pds/expectation.html#thm-lie
[c] https://morrison-lab.github.io/pds/variance-covariance.html#thm-var-lincom

Corollary 3.1 (Method-of-moments estimators of the variance components)**.** *Under Definition 2.1,* $\hat{\sigma}^2 \overset{def}{=} MSE$ *and*

$$\hat{\sigma}_\alpha^2 \overset{def}{=} \frac{MSA - MSE}{m} \tag{14}$$

are unbiased estimators of σ^2 *and* σ_α^2*.*

Proof

Proof. This proof uses Theorem 3.1.

$$\begin{aligned}
\mathrm{E}[\hat{\sigma}_\alpha^2] &= \mathrm{E}\left[\frac{\mathrm{MSA} - \mathrm{MSE}}{m}\right] && \text{(definition of the estimator)} \\
&= \frac{\mathrm{E}[\mathrm{MSA}] - \mathrm{E}[\mathrm{MSE}]}{m} && \text{(linearity of expectation)} \\
&= \frac{m\sigma_\alpha^2 + \sigma^2 - \sigma^2}{m} && \text{(expected mean squares)} \\
&= \frac{m\sigma_\alpha^2}{m} && \text{(cancel } \sigma^2\text{)} \\
&= \sigma_\alpha^2 && \text{(cancel } m\text{)}
\end{aligned} \tag{15}$$

$\mathrm{E}[\hat{\sigma}^2] = \mathrm{E}[\mathrm{MSE}] = \sigma^2$ directly from Theorem 3.1. □

Exm

Example 3.1 (Variance components for the dyestuff batches)**.** The ANOVA table for Example 2.1 gives the mean squares:

```
anova(lm(Yield ~ Batch, data = Dyestuff))
#> # A tibble: 2 x 5
#>      Df `Sum Sq` `Mean Sq` `F value` `Pr(>F)`
#>   <int>    <dbl>     <dbl>     <dbl>    <dbl>
#> 1     5   56357.    11271.      4.60  0.00440
#> 2    24   58830.     2451.     NA    NA
```

By Corollary 3.1, $\hat{\sigma}^2 = 2451.3$ and

$$\begin{aligned} \hat{\sigma}^2_\alpha &= \frac{11271.5 - 2451.3}{5} && \text{(substitute the mean squares and } m = 5) \\ &= \frac{8820.2}{5} && \text{(subtract)} \\ &= 1764.04 && \text{(divide)} \\ &\approx 1764.0 && \text{(round to one decimal place)} \end{aligned} \tag{16}$$

The `lmer()` function in the `lme4` package (Bates et al. 2015) fits the same model by restricted maximum likelihood (REML); `(1 | Batch)` requests a random intercept for each batch:

```
dyestuff_lmer <- lmer(Yield ~ 1 + (1 | Batch), data = Dyestuff)
VarCorr(dyestuff_lmer) |> as.data.frame()
#> # A tibble: 2 x 5
#>   grp      var1        var2   vcov sdcor
#>   <chr>    <chr>       <chr> <dbl> <dbl>
#> 1 Batch    (Intercept) <NA>  1764.  42.0
#> 2 Residual <NA>        <NA>  2451.  49.5
```

In this balanced design with a positive moment estimate, the REML estimates match the method-of-moments estimates.

The method-of-moments estimate $\hat{\sigma}^2_\alpha$ is negative whenever MSA $<$ MSE, which is impossible for a variance; likelihood-based methods such as REML constrain the estimate to be nonnegative, and they also handle unbalanced designs (Rocke 2021; Bates et al. 2015).

4 Nested random effects

A model can have more than one level of random effects. When each lower-level group belongs to exactly one higher-level group, the random effects are **nested** (Rocke 2021).

Definition 4.1 (Two-level nested random-effects model)**.** Suppose a laboratories each analyze b batches, with r replicate measurements per batch. The **two-level nested random-effects model** is

$$Y_{ijk} \overset{\text{def}}{=} \mu + L_i + B_{ij} + \epsilon_{ijk}, \tag{17}$$

where $L_i \sim_{\text{iid}} \mathrm{N}(0, \sigma^2_L)$ is the effect of laboratory i, $B_{ij} \sim_{\text{iid}} \mathrm{N}(0, \sigma^2_B)$ is the effect of batch j within laboratory i, $\epsilon_{ijk} \sim_{\text{iid}} \mathrm{N}(0, \sigma^2)$ is the replicate error, and all three sets of random variables are mutually independent.

By the same argument as Theorem 3.1, the expected mean squares of the nested ANOVA are $\mathrm{E}[\mathrm{MS}_L] = br\sigma^2_L + r\sigma^2_B + \sigma^2$, $\mathrm{E}[\mathrm{MS}_B] = r\sigma^2_B + \sigma^2$, and $\mathrm{E}[\mathrm{MS}_E] = \sigma^2$ (Rocke 2021). So the correct F test for laboratories divides MS_L by MS_B, not by MS_E.

Exm

Example 4.1 (A cooperative trial)**.** The `coop` data in the `MASS` package (Venables and Ripley 2002) come from a cooperative trial in which $a = 6$ laboratories each analyzed $b = 3$ batches, a month apart, with $r = 2$ replicates per batch. We use the measurements of specimen `S1`:

```
coop2 <- subset(MASS::coop, Spc == "S1")
anova(lm(Conc ~ Lab + Lab:Bat, data = coop2))
#> # A tibble: 3 x 5
#>      Df `Sum Sq` `Mean Sq` `F value`  `Pr(>F)`
#>   <int>    <dbl>     <dbl>     <dbl>     <dbl>
#> 1     5    1.89    0.378       60.0   1.35e-10
#> 2    12    0.204   0.0170       2.70  2.77e- 2
#> 3    18    0.113   0.00630     NA    NA
```

The `Lab:Bat` term is the batch-within-laboratory effect. The F statistic that `anova()` prints for `Lab` uses MS_E as its denominator, which is wrong here; the correct statistic is $F = 0.37804/0.01703 \approx 22.2$ on $(5, 12)$ degrees of freedom.
The equivalent mixed model has a random intercept for each laboratory and for each batch within each laboratory:

```
coop_lmer <- lmer(Conc ~ 1 + (1 | Lab) + (1 | Lab:Bat), data = coop2)
VarCorr(coop_lmer) |> as.data.frame()
#> # A tibble: 3 x 5
#>   grp      var1        var2     vcov  sdcor
#>   <chr>    <chr>       <chr>   <dbl>  <dbl>
#> 1 Lab:Bat  (Intercept) <NA>  0.00537 0.0733
#> 2 Lab      (Intercept) <NA>  0.0602  0.245
#> 3 Residual <NA>        <NA>  0.00630 0.0794
```

The estimated variance components are $\hat{\sigma}^2 \approx 0.0063$ (same-day replicates), $\hat{\sigma}^2_B \approx 0.0054$ (batches a month apart), and $\hat{\sigma}^2_L \approx 0.0602$ (laboratories). Most of the variation in the measurements comes from differences between laboratories.

Testing whether a variance component is zero is harder than testing a fixed effect, because $\sigma^2 = 0$ lies on the boundary of the parameter space: a variance cannot be negative, so the usual large-sample distribution of the likelihood ratio statistic does not apply (Rocke 2021).

5 Multilevel models: complete, no, and partial pooling

A **multilevel model** is a regression model whose coefficients are allowed to vary by group, with the group-level coefficients themselves given a model (Gelman and Hill 2007, chap. 12). The random-effects models in Definition 2.1 and Definition 4.1 are the simplest multilevel models.

Exm

Example 5.1 (Radon in Minnesota homes)**.** Gelman and Hill (2007, chap. 12) analyze measurements of indoor radon, a radioactive gas that can cause lung cancer, from 919 houses in 85 Minnesota counties. We use the copy of these data distributed with the `rstanarm` R package (Goodrich et al. 2024):

- `log_radon`: natural log of the radon measurement in the house;
- `floor`: `0` if the measurement was taken in the basement, `1` if on the first floor;
- `county`: the county the house is in;
- `log_uranium`: natural log of the county-level soil uranium measurement.

```
load(here::here("inst/extdata/radon.rda"))
radon_n <- table(radon$county)
summary(as.vector(radon_n))
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>     1.0     3.0     5.0    10.8    10.0   116.0
```

The number of houses per county varies from 1 to 116.

There are three ways to estimate the mean log radon level in each county (Gelman and Hill 2007, chap. 12):

- **Complete pooling** ignores the counties and estimates every county's mean by the overall mean.
- **No pooling** estimates each county's mean by its own sample mean $\bar{y}_j$, which is very noisy for counties with only one or two houses.
- **Partial pooling** fits the random-intercept model $Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$ for house i in county j (Definition 2.1, with the roles of the indices swapped and unequal group sizes n_j) and estimates each county's mean as a compromise between $\bar{y}_j$ and μ.

Theorem 5.1 (Partial-pooling estimate of a group mean)**.** *In the random-intercept model with known μ, σ_α^2, and σ^2, the predicted mean of group j, $\mathrm{E}\big[\mu + \alpha_j \mid \text{data}\big]$, is*

$$\hat{\mu}_j = w_j \bar{y}_j + (1 - w_j)\mu, \qquad w_j \overset{def}{=} \frac{n_j/\sigma^2}{n_j/\sigma^2 + 1/\sigma_\alpha^2}, \tag{18}$$

a weighted average of the group's sample mean and the overall mean, with weights proportional to their precisions (Gelman and Hill 2007, chap. 12).

The proof of Theorem 5.1 uses the conditional distribution of one normal variable given another, which is beyond the scope of this chapter. The weight w_j increases with the group size n_j: large groups keep estimates close to their own sample means, and small groups are pulled, or **shrunk**, toward the overall mean.

Exm

Example 5.2 (Shrinkage of county mean radon levels)**.** We fit the random-intercept model to the radon data from Example 5.1:

```
radon_lmer0 <- lmer(log_radon ~ 1 + (1 | county), data = radon)
VarCorr(radon_lmer0) |> as.data.frame()
#> # A tibble: 2 x 5
#>   grp      var1        var2    vcov sdcor
#>   <chr>    <chr>       <chr>  <dbl> <dbl>
#> 1 county   (Intercept) <NA>  0.0886 0.298
#> 2 Residual <NA>        <NA>  0.587  0.766
fixef(radon_lmer0)
#> (Intercept)
#>     1.34983
```

The estimates are $\hat{\mu} \approx 1.350$, $\hat{\sigma}_\alpha^2 \approx 0.0886$, and $\hat{\sigma}^2 \approx 0.587$. Aitkin County has $n_j = 4$ houses with sample mean $\bar{y}_j \approx 0.715$. Plugging these estimates into Theorem 5.1:

$$\begin{aligned}
w_j &= \frac{4/0.587}{4/0.587 + 1/0.0886} && \text{(substitute the estimates)} \\
&\approx \frac{6.81}{6.81 + 11.29} && \text{(divide)} \\
&\approx 0.377 && \text{(divide)} \\
\hat{\mu}_j &\approx 0.377 \times 0.715 + 0.623 \times 1.350 && \text{(substitute into the weighted average)} \\
&\approx 1.111 && \text{(multiply and add)}
\end{aligned} \tag{19}$$

Aitkin County's estimate moves most of the way from its own mean toward the overall mean. In contrast, Hennepin County has $n_j = 105$ houses, so its weight is about 0.94 and its estimate barely moves. The same calculation for every county reproduces the estimates that `lme4` reports:

```
radon_vc <- as.data.frame(VarCorr(radon_lmer0))$vcov
radon_w <- (radon_n / radon_vc[2]) /
  (radon_n / radon_vc[2] + 1 / radon_vc[1])
radon_ybar <- tapply(radon$log_radon, radon$county, mean)
radon_pp <- radon_w * radon_ybar + (1 - radon_w) * fixef(radon_lmer0)
max(abs(radon_pp - coef(radon_lmer0)$county[, 1]))
#> [1] 3.77476e-15
```

```
plot(
  as.vector(radon_n), radon_ybar,
  log = "x", pch = 1,
  xlab = "Houses sampled in county (log scale)",
  ylab = "Estimated county mean log radon"
)
points(as.vector(radon_n), radon_pp, pch = 19)
abline(h = fixef(radon_lmer0), lty = 2)
legend(
  "topright",
  legend = c("No pooling", "Partial pooling"),
  pch = c(1, 19)
)
```

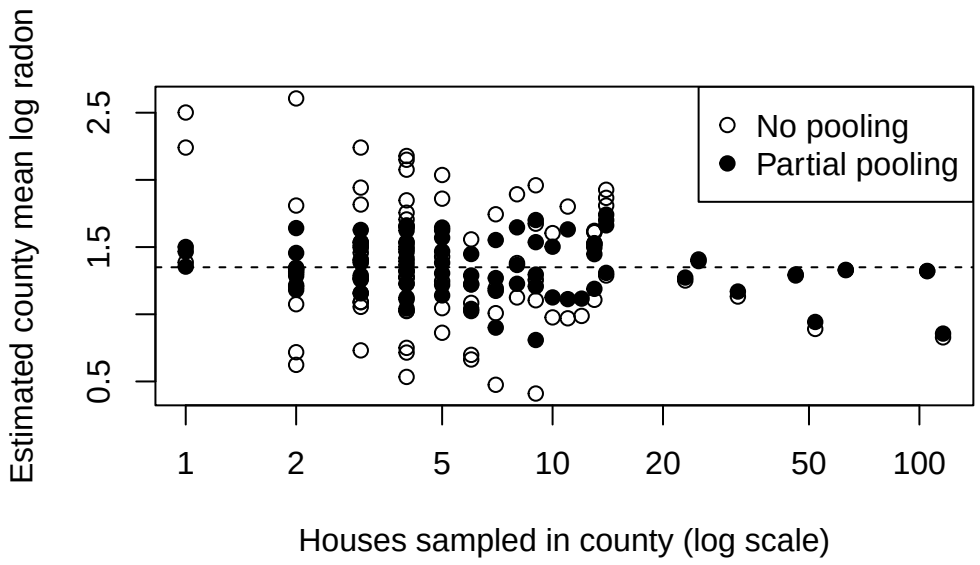


Figure 1: No-pooling and partial-pooling estimates of the county mean log radon levels; the dashed line is the complete-pooling estimate.

6 Adding predictors to a multilevel model

Covariates enter a multilevel model as fixed effects, and their coefficients can also vary by group (Gelman and Hill 2007, chap. 12). For house i in county j with floor indicator x_{ij}:

- the **varying-intercept** model is $Y_{ij} = \beta_0 + \alpha_j + \beta_1 x_{ij} + \epsilon_{ij}$;
- the **varying-intercept, varying-slope** model is $Y_{ij} = \beta_0 + \alpha_j + (\beta_1 + \gamma_j) x_{ij} + \epsilon_{ij}$, where each county's pair (α_j, γ_j) is bivariate normal with mean zero.

Exm

Example 6.1 (Radon by floor of measurement)**.**

```
radon_lmer1 <- lmer(log_radon ~ floor + (1 | county), data = radon)
radon_lmer2 <- lmer(log_radon ~ floor + (1 + floor | county), data = radon)
fixef(radon_lmer1)
#> (Intercept)       floor
#>    1.492393   -0.662889
VarCorr(radon_lmer2)
#>  Groups   Name        Std.Dev. Corr
#>  county   (Intercept) 0.3371
#>           floor       0.3285   -0.37
#>  Residual             0.7168
```

In the varying-intercept model, measurements on the first floor average about 0.66 lower on the log scale than basement measurements in the same county, a ratio of $\exp\{(\} - 0.66) \approx 0.52$. The varying-slope model lets that difference vary by county.
To compare the two models, `anova()` refits both by maximum likelihood and computes a likelihood ratio test:

```
anova(radon_lmer1, radon_lmer2)
#> # A tibble: 2 x 8
#>    npar   AIC   BIC logLik `-2*log(L)` Chisq    Df `Pr(>Chisq)`
#>   <dbl> <dbl> <dbl>  <dbl>       <dbl> <dbl> <dbl>        <dbl>
#> 1     4 2098. 2117. -1045.       2090. NA       NA       NA
#> 2     6 2099. 2128. -1044.       2087.  2.66     2        0.264
```

The varying-slope model adds two parameters (the slope variance and its correlation with the intercept), and the likelihood ratio test gives $p \approx 0.26$. Because the null value of the slope variance, zero, is on the boundary of the parameter space, this p-value is conservative (Rocke 2021); AIC and BIC also favor the simpler model.

Exm

Example 6.2 (A county-level predictor)**.** Soil uranium is measured once per county, so it is a **group-level predictor**. Adding it to the varying-intercept model:

```
radon_lmer3 <- lmer(
  log_radon ~ floor + log_uranium + (1 | county),
  data = radon
)
fixef(radon_lmer3)
#> (Intercept)       floor log_uranium
#>    1.496157   -0.638778    0.697499
VarCorr(radon_lmer3) |> as.data.frame()
#> # A tibble: 2 x 5
#>   grp      var1        var2    vcov sdcor
#>   <chr>    <chr>       <chr>  <dbl> <dbl>
#> 1 county   (Intercept) <NA>  0.0218 0.148
#> 2 Residual <NA>        <NA>  0.531  0.728
```

The estimated between-county variance falls from about 0.099 without uranium to about 0.022 with it, while the within-county variance barely changes: county uranium explains most of the differences between counties.

7 Multilevel logistic regression

Random effects can be added to any generalized linear model. A **generalized linear mixed model** (GLMM) adds random effects to the linear predictor, so for a binary outcome with random

intercepts,

$$\log\left\{\frac{\pi_{ij}}{1-\pi_{ij}}\right\} = \tilde{x}_{ij}^{\top}\tilde{\beta} + \alpha_j, \qquad \alpha_j \sim_{\text{iid}} \mathrm{N}(0, \sigma_\alpha^2) \tag{20}$$

(Dobson and Barnett 2018, chap. 11; Gelman and Hill 2007, chap. 14). Each group's odds are multiplied by $\exp\{\alpha_j\}$, so about 95% of groups have odds within a factor of $\exp\{\pm 2\sigma_\alpha\}$ of the odds for a typical group (Rocke 2021).

Exm

Example 7.1 (Contagious bovine pleuropneumonia)**.** The `cbpp` data in the `lme4` package (Bates et al. 2015) record the number of new cases of contagious bovine pleuropneumonia in 15 herds of zebu cattle in Ethiopia, each observed in up to four periods. The herds are a sample of all herds, so we model herd as a random effect and period as a fixed effect:

```
cbpp_glmer <- glmer(
  cbind(incidence, size - incidence) ~ period + (1 | herd),
  family = binomial,
  data = cbpp
)
summary(cbpp_glmer)$coefficients
#>              Estimate Std. Error  z value    Pr(>|z|)
#> (Intercept) -1.398343   0.231214 -6.04783 1.46810e-09
#> period2     -0.991925   0.303151 -3.27205 1.06769e-03
#> period3     -1.128216   0.322830 -3.49477 4.74473e-04
#> period4     -1.579745   0.422049 -3.74304 1.81808e-04
VarCorr(cbpp_glmer)
#>  Groups Name        Std.Dev.
#>  herd   (Intercept) 0.6421
```

The estimated herd standard deviation is $\hat{\sigma}_\alpha \approx 0.642$, so about 95% of herds have odds of a new case between $\exp\{-2 \times 0.642\} \approx 0.28$ and $\exp\{2 \times 0.642\} \approx 3.6$ times those of a typical herd. Holding herd fixed, the odds of a new case in period 4 are $\exp\{-1.58\} \approx 0.21$ times the odds in period 1.

The fixed-effect coefficients in a GLMM are **conditional** on the random effect: they compare odds within the same group. Because the logit link is nonlinear, these coefficients are not the same as population-averaged odds ratios (Dobson and Barnett 2018, chap. 11).

8 Further reading

For more, see:

- David Rocke[3]'s materials from the 2021 edition of this course[4]
  - May 25 - June 1 lectures
- Other UC Davis courses:
  - EVE 225[5]: "Linear Mixed Modeling in Ecology & Evolution"
    - usually taught every other winter or spring by Kate Laskowski[6]
    - materials, including syllabus and lecture videos: https://laskowskilab.faculty.ucdavis.edu/teaching-2/
  - STA/BST 224[7]: "Analysis of Longitudinal Data"
    - usually taught every spring by Shuai Chen[8]

[3] https://dmrocke.ucdavis.edu/
[4] https://dmrocke.ucdavis.edu/Class/EPI204-Spring-2021/EPI204-Spring-2021.html
[5] https://catalog.ucdavis.edu/search/?q=EVE+225
[6] https://eve.ucdavis.edu/people/kate-laskowski
[7] https://catalog.ucdavis.edu/search/?P=BST%20224
[8] https://shuaichen.weebly.com/

    - should be accessible after completing Epi 204
  - EPI 226[9] "Methods for Longitudinal & Repeated Measurement Data"
    - usually taught by Heejung Bang[10]
  - PSC 205D[11] "Multilevel Models"
  - PSC 205G[12] "Applied Longitudinal Data Analysis"
  - STA 101[13] "Advanced Applied Statistics for the Biological Sciences"
  - STA 207[14] "Statistical Methods for Research II"
  - STA 232B[15] "Applied Statistics II"
    - usually taught every winter by Jiming Jiang[16]
  - PLS 207[17]: "Applied Statistical Modeling for the Environmental Sciences"
  - EDU 236[18]: "Application of Hierarchical Linear Models in Education Research"
  - HDE 205[19]: "Longitudinal Data Analysis"
- Books:
  - Dobson and Barnett (2018) Chapter 11[20]
  - Vittinghoff et al. (2012) Chapter 7[21]
  - Gelman and Hill (2007)
  - Jiang and Nguyen (2021)
    - by UC Davis Statistics Professor and GGE faculty member Jiming Jiang[22]
  - Faraway (2016)
  - McCulloch et al. (2008)
  - Hedeker and Gibbons (2006)
  - Wakefield (2013)
  - Zuur (2009)
  - Diggle et al. (2013)
  - Fitzmaurice et al. (2012)
  - Fitzmaurice et al. (2009)
  - Gałecki and Burzykowski (2013)
  - Congdon (2020)
  - Molenberghs and Verbeke (2005)
  - Verbeke and Molenberghs (2000)
  - Jewell and Hubbard (2016)
    - by UC Berkeley professors

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[20] https://www.taylorfrancis.com/chapters/mono/10.1201/9781315182780-11/clustered-longitudinal-data-annette-dobson-adrian-barnett?context=ubx&refId=95f6c50e-093a-4488-a042-92a9f151a4b5
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[22] https://www.stat.ucdavis.edu/~jiang/

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